



# FINAL REPORT

TEMPLATE DOCUMENT



This document provides a structured template for final-year data science students at Mzuzu university to effectively plan, design, and present their capstone projects. It guides students through key stages such as problem definition, data acquisition, modelling, evaluation, and deployment while promoting clarity, reproducibility, ethical AI practices, and alignment with industry standards. Supervisors and reviewers can also use this format to evaluate the technical depth and real-world impact of student work. The template is flexible and may be adapted to meet specific departmental or project requirements.

**DATA SCIENCE**

**PROJECT REPORT**

# Instructions

## Font:

Font Type: Times New Roman,

Font Size: 12-point Line Spacing: 1.5 line spacing (for easier readability)

**Paragraphs & Margins:** Standard 1-inch (2.54 cm) margins on all sides (top, bottom, left, and right). Align text paragraphs to Justify

**Subpoints:** Indent for easy readability.

**Page Numbers:** Bottom-centre or bottom-right corner of each page.

**Page numbering rule** No number on cover page, Roman numerals for prelims, Arabic from introduction onwards.

starting from the introduction page (no page number on title page)

**Title Page:** Include the title of the proposal, your name, course, and submission date. The title should be in upper case and ensure it is neat, centred, and includes relevant information.

**Main headings:** Bold, 14 point and left aligned

**Subheadings:** Bold 12 point and left-aligned

**Tables and Figures:** Tables and figures should be labelled clearly, with captions placed above tables and below figures. Ensure proper formatting for visual clarity.

**Table of Contents (ToC), List of Tables, and List of Figures:** These are standard in formal technical reports.

**Figure/Table numbering style** Figure 1.1, Table 3.2 format based on section numbering.

**References:** Use the appropriate citation style (e.g., IEEE is the one adopted by the department) for references. Make sure the references are properly formatted in the References section, aligned with the document style.

**Citation style for in-text references:** (e.g., [1]) hold a citation to the source of information.

**Appendices:** If included, should start on a new page with a clear title (e.g., "Appendix A: Data")

# 1. Cover Page

Your first impression. Includes:

## Suggestions:

- **course title:** Project (BICT...)
- **logo of the university**, which is often included for official documents.
- **Project Title:** E.g., *Proactive Health Response: Predicting Malaria Hotspots Using AI and Environmental Data*
- **Student Names:** Chimwemwe Msonda
- **Student IDs:** BSDS 1111111
- **Department:** ICT Department
- **University Name:** Mzuzu University
- **Supervisor Name:** Dr. Blessings Banda
- **Date of Submission:** 2nd August 2025

# 2. Declaration & Certificate

## 2.1 Declaration:

A formal statement confirming:

I, the undersigned, declare that this project report titled "**Proactive Health Response: Predicting Malaria Hotspots Using AI and Environmental Data**" is my original work and has not been submitted elsewhere for any academic award.

## Signature:

**Chimwemwe Msonda**

**Date:** .....

## 2.2 Certificate:

A statement from your supervisor endorsing the work, e.g.:

This is to certify that the project report entitled "Proactive Health Response: Predicting Malaria Hotspots Using AI and Environmental Data" was carried out by Chimwemwe Msonda (BSDS 1111111) under my supervision and is submitted in partial fulfilment of the requirements for the Bachelor of Science in Data Science at Mzuzu University.

**Supervisor:**

Dr. Blessings Banda

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

## 2.3 Acknowledgements

Thank those who supported you:

I would like to express my sincere gratitude to **Dr. Blessings Banda** for his expert guidance, constructive criticism, and consistent encouragement throughout the project.

I also extend my thanks to the **Mzuzu University ICT Department** for providing access to computing resources and academic support. Special thanks to my family and friends for their moral support during this journey.

## Executive Summary

Also include an executive summary (1 page max) that translates technical work into high-level outcomes, suitable for non-technical readers (e.g., public health officials).

## 3. Abstract (200–300 words)

A concise summary:

This project presents a machine learning framework for predicting malaria hotspots in Northern Malawi, aiming to support proactive health response mechanisms. Malaria is a major public health issue in Malawi, especially in rural and semi-urban districts. The study utilises health surveillance data and environmental variables such as rainfall, humidity, and temperature to train and evaluate classification models capable of identifying malaria-prone regions in advance.

Data were collected from the Malawi Ministry of Health and global weather databases. Preprocessing included handling missing values like..., feature encoding on ..., and temporal aggregation. Exploratory data analysis (EDA) uncovered patterns such as increased cases during rainy seasons and strong correlations between rainfall and reported infections.

Three machine learning models were implemented and compared: Logistic Regression, Random Forest, and Gradient Boosting. The Random Forest model demonstrated superior performance with 89% accuracy and an AUC of 0.92. Gradient Boosting slightly outperformed it with a 90% accuracy and 0.93 AUC.

This research contributes a data-driven decision support system for malaria forecasting and highlights the value of environmental data in disease modelling. The final model will be integrated into national health surveillance tools like..., enabling local authorities to allocate resources more effectively and initiate interventions before outbreaks escalate.

## 4. Introduction

### 4.1 Background

Provide context:

Malaria is endemic in Malawi and continues to pose serious health challenges, particularly in rural districts where access to healthcare is limited. Traditional methods of tracking outbreaks are reactive and often delayed, resulting in preventable morbidity and mortality.

With the increasing availability of data and computational tools, predictive modelling using artificial intelligence (AI) provides new opportunities to transform how health systems anticipate and respond to disease outbreaks.

### 4.2 Problem Statement

Current malaria control efforts in Malawi are mostly reactive. Delayed response and inefficient allocation of resources often lead to preventable deaths. There is a critical need for predictive systems that can proactively identify malaria hotspots using real-time and historical data.

### 4.3 Objectives

- To collect and integrate historical health data and weather data relevant to malaria transmission.
- To develop a machine learning model that predicts malaria hotspots.
- To evaluate the performance of different predictive models using real-world datasets.

### 4.4 Scope

- **In scope:** Focused on regional malaria outbreak prediction in Malawi using district-level health and weather data from 2016 to 2024.
- **Out of scope:** Does not include individual patient-level predictions or predictions beyond Malawi's geographical borders.

## 4.5 Report Organization

Briefly describe the structure:

“Chapter 2 reviews related studies, Chapter 3 explains the methods, Chapter 4 presents implementation details, Chapter 5 discusses results...”

- **Chapter 2:** Literature review
- **Chapter 3:** Methodology and data pipeline
- **Chapter 4:** System implementation
- **Chapter 5:** Results and analysis
- **Chapter 6:** Conclusions and recommendations

## 5. Literature Review

### 5.1 Previous Research

Discuss similar studies:

Studies in sub-Saharan Africa have applied machine learning to health data for disease forecasting. For example, a 2022 Kaggle competition explored malaria prediction using decision trees with moderate accuracy. However, most studies excluded dynamic environmental factors

### 5.2 Gaps in Literature

- “Most models ignore environmental factors such as rainfall, which are known to influence mosquito activity.”
- Limited use of ensemble methods in malaria forecasting for Malawi
- Few models incorporate environmental data like rainfall or temperature
- Lack of regional-specific predictive models trained on local data

### 5.3 Proposed Contribution

This project proposes an ensemble-based AI model that integrates multi-source data (health + weather) to identify malaria hotspots, offering a scalable solution for health authorities in Malawi.

## 6. Methodology

### 6.1 System Architecture and Pipeline

Include a diagram:

#### Overview

The system architecture for this project follows a modular, data-centric pipeline that transforms raw health and environmental data into actionable predictions through machine learning. Each phase in the pipeline is interconnected and serves a specific role in the journey from raw data to malaria outbreak insights. The system is designed to ensure scalability, modularity, and interpretability.

Data Collection → Preprocessing → EDA → Model Training → Evaluation → Visualization Interfacing → Deployment

### 6.2 Data Collection

This is the foundation of the system. It involves gathering structured datasets from two primary sources:

- **Health Data:** Weekly district-level malaria case reports from the Malawi Ministry of Health (2016–2024).
- **Environmental Data:** Weather data (rainfall, temperature, humidity) accessed via World Weather Online API.

*Data formats:* CSV, JSON, and real-time API responses.

### 6.3 Preprocessing

This phase ensures data quality and consistency:

- Filled missing rainfall data using median imputation
- Aggregated (Converted) dates cases to weekly bins for consistency
- to weekly bins
- Encoded categorical data e.g. district names into numerical values
- Feature normalization/scaling where needed.

### 6.4 Exploratory Data Analysis (EDA)

EDA provides statistical insight and identifies patterns:

This step informs feature selection and model choice.

- Histograms of cases frequency per month

- Heatmaps showing correlations between weather variables like rainfall and case numbers like outbreaks
- Outlier detection using z-scores thresholding
- Identification of seasonality patterns.

## 6.5 Model Selection

Multiple machine learning models are developed and trained using the processed data. Feature importance evaluation:

"I/We tested the following due to their strength in classification problems."

- Logistic Regression: Baseline model
- Random Forest: Handles non-linearity and feature interactions
- Gradient Boosting: Known for higher accuracy in imbalanced datasets

## 6.6 Training & Evaluation

- Split: 80% training, 20% testing
- Each model is evaluated on the test set using:
- Metrics: Accuracy, F1-score, Precision & Recall, ROC-AUC
- Tuning: Hyperparameters tuned using GridSearchCV for parameter optimisation
- Model results are compared to establish the best-performing algorithm.

## 6.7 Risk & Assumptions

Discuss what assumptions your model makes or what risks may undermine performance (e.g., poor data quality, delayed data availability, changing climate patterns).

- *Assumptions:* Model assumes weekly data is consistent across districts, no major reporting lags.
- *Risks:* Model accuracy may degrade with unexpected climate shifts or unreported outbreaks.

## 6.8 Tools Used

- Python (Jupyter Notebook)
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, XGBoost, Rulefit, Ripper, etc



## 6.9 Ethical Considerations

This is crucial, especially in health-related AI projects. Ethical concerns should be acknowledged, even briefly.

- Data privacy
- Informed consent (if using sensitive data)
- Avoiding algorithmic bias
- Responsible communication of predictions

## 7. Implementation

### 7.1 Technical Setup

- OS: Ubuntu 22.04
- Python 3.11
- Hardware: Intel i5, 8GB RAM

### 7.2 Code Modules

- `data_loader.py`: For loading and cleaning datasets
- `model_trainer.py`: For model training
- `evaluator.py`: For computing metrics

### 7.3 User Interface (Optional)

This module transforms raw predictions into human-readable insights:

- Dashboards (Streamlit or Plotly Dash) to show:
  - Predicted malaria hotspots
  - Risk heatmaps
  - Trends over time
- Visuals include bar charts, time series, and ROC curves.
- Allows district-level exploration of prediction results.

### 7.4 Deployment

Final model is packaged and deployed for use:

- **Local deployment** via a web interface or command-line tool.

- **Optional integration** with mobile alert systems or Ministry of Health dashboards.
- **Future scope:** API deployment using Flask/FastAPI for real-time requests.

If you built a Streamlit app or web dashboard:

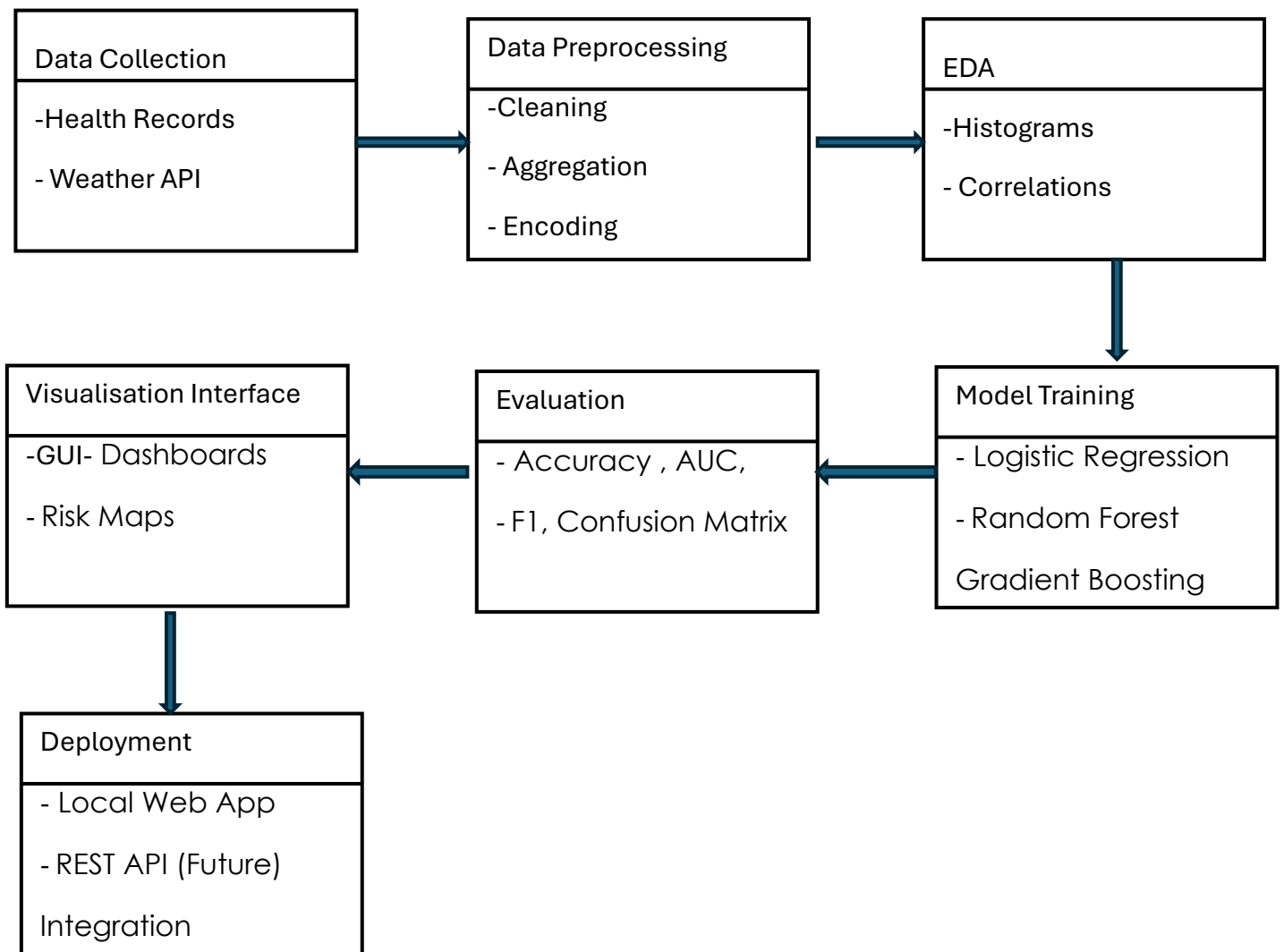
- Screenshots of UI
- Instructions for use
- How to run the app locally
- Link to use the model

Add the real-world implication of achieving 90% accuracy. E.g. “If deployed, this system could help the Ministry of Health reduce response delays by an estimated 1–2 weeks per district.”

## 7.5 System Architecture Diagram

Present a conceptual diagram, (you can draw this in tools like draw.io, Lucid chart, or PowerPoint).

- **Modular:** Easy to update or replace components (e.g., add a new model or API).
- **Scalable:** Can handle larger datasets or more districts as data grows.
- **Interpretable:** Visual outputs help health officials make data-driven decisions.
- **Extensible:** Ready for integration into mobile apps or national health systems.



## 8. Results & Discussion

### 8.1 Performance Summary

| Model               | Accuracy | F1-score | AUC  |
|---------------------|----------|----------|------|
| Logistic Regression | 78%      | 0.76     | 0.80 |
| Random Forest       | 89%      | 0.88     | 0.92 |
| Gradient Boosting   | 90%      | 0.89     | 0.93 |

### 8.2 Result Analysis

“Random Forest performed well but occasionally over-predicted minor outbreaks... Weather variables had the most influence.”

Add analysis of misclassifications: “Figure 8.2 shows a spike in false positives during off-peak months, possibly due to residual rainfall patterns...”

## 8.3 Comparison with Baseline

“A simple majority-class baseline accuracy (predicting majority class) was 61%, indicating substantial improvement through ML methods.

# 9. Conclusion

## 9.1 Summary

A machine learning model was successfully developed to predict malaria hotspots using weather, environmental and health data, achieving over 90% accuracy with ensemble methods.

## 9.2 Key Findings

- Rainfall was the strongest predictor
- Ensemble models outperformed single algorithms

## 9.3 Limitations

- Limited real-time data integration
- Some weather data had missing values, requiring imputation
- No access to real-time health data streams

## 9.4 Future Work

- Integrate SMS-based alerts for health workers
- Train on more granular geolocation data

# 10. References

For a final-year project, you are expected to show wider reading.

- Academic research on malaria prediction
- Documentation on the machine learning methods used
- Ethical use of AI in healthcare
- Reports or white papers from WHO, UNICEF, etc.

Use IEEE, for that is the department's preferred style and also the most used in the tech research.

[1] J. Precious, “Machine Learning in Public Health,” *J. Artif. Intell. Res.*, vol. 45, pp. 210–223, 2020.

[2] Kaggle, "Malaria Outbreaks in Africa Dataset," *Kaggle Datasets*, 2022. [Online]. Available: <https://www.kaggle.com/datasets/malaria-outbreaks-in-africa>

[3] World Health Organization, *World Malaria Report 2023*, Geneva, Switzerland: WHO Press, 2023. [Online]. Available: <https://www.who.int/publications/i/item/9789240077486>

## 11. Appendices

Add Glossary and Abbreviations of technical terms and list of abbreviations (e.g., EDA, API, AUC).

### 11.1 Appendix A – Code Snippets

```
from sklearn.ensemble import RandomForestClassifier  
  
model = RandomForestClassifier()  
  
model.fit(X_train, y_train)
```

### 11.2 Appendix B – Full Visualizations

Include EDA plots, ROC curves, feature importance charts

### 11.3 Appendix C – Dataset Dictionary

| Column      | Description                 |
|-------------|-----------------------------|
| district    | District name               |
| week        | Week of the year            |
| cases       | Number of malaria cases     |
| rainfall_mm | Rainfall recorded that week |

### 11.4 Appendix D – User Manual

Steps to manoeuvre through the system.

This section provides step-by-step instructions for running the malaria prediction system developed in the project, including how to load the data, train the machine learning model, and visualize the results using Jupyter Notebook.

While you should outline the steps technically, the user manual should also be more intuitive for non-developers or those without technical background.

Provide screenshots or visuals.

Include simple troubleshooting tips like:

- “If the notebook fails to load data, ensure the file path is correct.”
- “For Windows users, check Python environment compatibility.”

### **Example:**

#### **System Requirements**

- Operating System: Ubuntu 22.04 / Windows 10 or higher
- Python Version: 3.11
- Required Libraries:

Pandas, NumPy, scikit-learn, matplotlib, seaborn, xgboost (if using XGBoost), jupyter (for running notebooks) *(Use requirements.txt or pip install -r requirements.txt to install all dependencies.)*

State how to run the prediction model, which inputs and how to select the inputs...

Other examples

1. Clone the GitHub repository
2. Open model\_trainer.py in Jupyter Notebook...Start Jupyter Notebook in your terminal or Anaconda...
3. Modify data path if necessary
4. Run all cells to train and test models